

DATA STRUCTURES AND ALGORITHMS using Java

Course Introduction

Objectives

- Instructor introduction
- Course description
- Text book(s) & Reference Recourses
- Learning Tools
- Requirements of the course
- Grading policy
- Academic policy

Instructor introduction

Name: Phan Đăng Cầu

Contacts:

- Email: caupd@fpt.edu.vn
- Mob: 0912 037 064

Course description

- This course introduce the fundamental concepts of data structures and the algorithms that proceed from them. Topics include the basics of algorithmic analysis, fundamental data structures (including stacks, queues, linked lists, hash tables, trees), recursion, and some important applications of these data stuctures and algorithms.
- Such data structures and algorithms as being implemented in Java are also given in this course.

Text book(s) & Reference Recourses

- Main books/resources:
- 1) Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser: Data Structures and Algorithms in Java, 6th Edition, 2014 (ebook)
- 2) Link to the book:
<http://coltech.vnu.edu.vn/~sonpb/DSA/Data%20Structures%20and%20Algorithms%20in%20Java,%206th%20Edition,%202014.pdf>
- 3) FU slides (ppt)
- 4) FU exercises (pdf)
- 5) Code files for students (java files)
- 6) FU CMS at <http://cms.fpt.edu.vn>.

Learning Tools

- JDK 1.7
- JDK 1.7 Documentation
- NetBeans 7.x

Requirements of the course

- Following lessons in classrooms
- Reading textbooks at home
- Completing workshops in time
- Completing and submitting assignment in time
- Discussing actively in your teams and in classrooms
- Presenting your presentations in classroom

Grading policy

- **On-going assessment:**
 - 2 Assignments (AS): 20%
 - 2 Progress tests (PT): 20%
- **Practical and Final Exams:**
 - 1 Practical Exam (PE): 30%
 - 1 Final Exam (FE): 30%
- **Total score (TS)** = $0.2 * AS + 0.2 * PT + 0.3 * PE + 0.3 * FE$
- **Completion Criteria:**
 - 1) Every on-going assessment (average) component > 0
 - 2) PE > 0 (no resit)
 - 3) FE >= 4 & TS >= 5

FPT-University Academic policy

Cheating, plagiarism and breach of copyright are serious offenses under this Policy.

- Cheating
 - Cheating during a test or exam is construed as talking, peeking at another student's paper or any other clandestine method of transmitting information.
- Plagiarism
 - Plagiarism is using the work of others without citing it; that is, holding the work of others out as your own work.
- Breach of Copyright
 - If you photocopy a textbook without the copyright holder's permission, you violate copyright law.

Assignments submission guide

1. Create the folder with a name like
login_name_ASX, e.g. 01245_HungNV_AS1 (1)
to contain your assignment.
2. Compress the whole folder (1) with the same name and
submit to cms.

OOP notion conventions

- Class name always start with uppercase character.
- Variable names and function names always start with the lowercase character.
- If a name consists of many words, from second words start all words by uppercase character.

examples for class names:

Rectangle, SecondDegreeEquation

examples for variable and function names:

sideOfRectangle, setDataToSafety